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Efficacy of Virtual Reality-Based Interventions in Rehabilitation Management of Breast Cancer Survivors: Protocol of a systematic review and meta-analysis

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Keywords:	Information technology < BIOTECHNOLOGY & BIOINFORMATICS, Breast tumours < ONCOLOGY, Rehabilitation medicine < INTERNAL MEDICINE

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Efficacy of Virtual Reality-Based Interventions in Rehabilitation Management of Breast Cancer Survivors: Protocol of a systematic review and meta-analysis

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Abstract

Introduction

Breast cancer is the leading cause of global cancer incidence and represents 11.7% of all cancer new cases. However, survivors suffer from a large number of intense physical and psychological symptoms, function deflicts, and adverse effects during and after the treatment, which greatly affects their quality of life. Virtual reality (VR) technology offers new and noninvasive approaches for different health care applications. This paper reports on the protocol for a systematic review and meta-analysis exploring the efficacy of VR-based interventions on rehabilitation management of breast cancer survivors (BCS).

Methods and analysis

This systematic review and meta-analysis protocol was prepared according to the recommendations from Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. Web of Science, PubMed, Embase, CINAHL, Cochrane Central Register of Controlled Trials, CNKI, Wanfang, VIP, and SinoMed will be used to search. This search will include randomized controlled trials, quasi-experimental studies, case-controlled trials, cohort studies published in English and Chinese. Further, the risk of bias for the studies included in the systematic review or meta-analysis will be assessed by the Cochrane risk-of-bias tool and quality assessment tool of JBI evidence-based health center. For the meta-analysis, the statistical program Review Manager V.5.3 will be used. The I^2 test will be used to determine the statistical heterogeneity among included studies.

Ethics and dissemination

Ethics approval will not be needed because the data used in this systematic review will be extracted from published studies. The results of the systematic review focusing on whether VR-based interventions are effective in the rehabilitation management of BCS. And it will be disseminated by publication in a peer-reviewed journal.

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Trial registration number

PROSPERO CRD42021250727

Keywords

Breast cancer survivors; virtual reality; rehabilitation management; systematic review; meta-analysis

Strengths and limitations of this study

- To the best of our knowledge, this is the first systematic review and meta-analysis to synthesize the totality of the evidence regarding VR-based interventions in rehabilitation management of BCS, which may provide scientific information regarding the efficacy of VR-Based interventions in rehabilitation management of BCS.
- The results of this meta-analysis will be helpful for clinicians to define rehabilitation strategies for improving rehabilitation effectiveness in BCS.
- This study will be limited to the English and Chinese languages, which may result in exclusions of potentially relevant studies published in other languages.
- There is a possibility that we get a small sample size and a limited number of studies, which may influence the validity and reliability of the findings.
- Despite these limitations, the findings of this systematic review and meta-analysis may suggest whether VR-based interventions are more efficacy in improving the rehabilitation management of BCS.

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Introduction

Female breast cancer has now surpassed lung cancer as the leading cause of global cancer incidence in 2020, with an estimated 2.3 million new cases and 685,000 deaths, respectively representing 11.7% of all cancer new cases and 6.9% of death cases.[1] Owning to the impact of screening programs, early detection, and accessible health care service, as well as advanced therapeutic strategies, the 5-year relative survival rate for individuals with breast cancer increased up to 82%.[2 3] However, survivors suffer from a large number of intense physical and psychological symptoms (i.e., pain, fatigue, depressive symptoms, anxiety), function deflicts (i.e., cognitive), and adverse effects (i.e., chemotherapy-induced nausea and vomiting) during and after the treatment, which greatly affect their quality of life.[4 5]

Emotional distress mainly includes fatigue, pain, anxiety, and depression, which is commonplace in cancer populations.[6] It was designated as the sixth vital sign in 2005 in Canada[7] and associated with a reduction in overall quality of life among cancer patients[8]. Cancer-related fatigue (CRF) is defined as a distressing, persistent, subjective sense of physical, emotional, and/or cognitive tiredness or exhaustion related to cancer or cancer treatment that is not proportional to recent activity and interferes with usual functioning.[9] Its levels peaked during breast cancer therapy and the prevalence of chronic fatigue increased the most after treatment.[10] Prevalence rates of severe fatigue ranged from 7% to 52%, and the pooled prevalence was 26.9%. Risk factors associated with fatigue were survivors treated with surgery, radiotherapy, and chemotherapy, hormone therapy, and survivors without a partner.[11] Patients with anemia are prone to the symptom of fatigue owning to the lower hemoglobin level after chemotherapy.[12] The prevalence of symptoms of depression varied from 9.4% to 66.1% and of anxiety from 17.9% to 33.3%.[13] Factors associated with the risk of anxiety and depression were age, place of residence, marital status, educational level, religion, stage of cancer, and current activity-burden of symptoms.[14] Anxiety was not more prevalent among women with early-stage breast cancer.[13] There is an increased risk of depression for BCS at one year after diagnosis and decreases over the ensuing

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years.[13] The prevalence of persistent post-surgical pain in BCS ranged from 2% to 78%.[15] Younger age, radiotherapy, axillary lymph node dissection, and greater acute postoperative pain were associated with persistent pain after breast cancer surgery[16]. Those who conducted axillary lymph node dissection are with an absolute increase in the risk of persistent pain of 21%.[16]

Cancer-related cognitive impairment (CRCI) is characterized as deficits in areas of cognition including memory, attention, information processing speed, and executive function.[17] [18] Between 15% and 50% of individuals with breast cancer receiving chemotherapy experience persisting cognitive impairment,[19] often referred to as "chemobrain".[20] The duration of symptoms may extend years after the completion of treatment.[17] BCS are at lifelong risk for the development of breast cancer-related lymphedema (BCRL)[21] and the incidence is 21.4%[22]. There is strong evidence showing that higher body mass index (BMI), the larger number of dissected nodes, certain chemotherapy agents (e.g., taxane-based regimen), the extent of surgery (e.g., total mastectomy), larger irradiation field, and sedentary lifestyles were associated with BCRL.[22-24] Due to the disruption of the lymphatic system after surgery or radiation treatment that resulted in the accumulation of lymph fluid that leads to BCRL.[25] BCRL is a chronic, potentially debilitating condition that involves progressive swelling, limited range of motion, and causing feelings of pain, numbness, which requires life-long symptom management.[21]

The development of VR offers new and noninvasive approaches for different health care applications. It creates interactive computer-generated worlds by visual, listening, and touch simulation, where an individual feels being. VR as a distraction tool improves cancer patients' emotional well-being and decreases cancer-related psychological symptoms, which improves their quality of life.[26] VR-based rehabilitation system has been identified to be effective for patients with weak arms and diminished ability to grasp.[27] It uses VR to engage the patient in upper body bimanual exercises and simultaneously to provide cognitive training and affective

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relief for the patient. However, it is unclear whether VR-based interventions could promote the rehabilitation management of BCS. Additionally, up to now, no systematic review and meta-analysis is investigating the association between VR and rehabilitation management of BCS. So, we will qualitatively and quantitatively examine the effects of VR-based interventions on BCS in this systematic review and meta-analysis.

Objectives

This systematic review and meta-analysis aimed to report on the studies using VR-based interventions for rehabilitation management in BCS and to quantitatively evaluate the efficacy of VR-based intervention in breast cancer-related rehabilitation management. The findings of this systematic review and meta-analysis can provide direction for future research.

Methods and analysis

Study design and registration

This systematic review and meta-analysis protocol was prepared according to the recommendations from Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement[28] and registered on PROSPERO (Registration number: CRD42021250727).

Eligibility criteria

Type of studies

We will include all randomized controlled trials, quasi-experimental studies, case-controlled trials, cohort studies.

Types of participants/population

We will include all trials which have included patients after first or recurrent breast cancer, regardless of demographic age, race, and education status.

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Types of interventions

We will select all trials assessing VR-based interventions whatever it may be and whatever its aim. The selection process will not be based on the type or the nature of the VR-based interventions (immersive and non-immersive virtual environment) in trials.

Types of outcome measures

Symptoms of pain, depression, anxiety; cognitive function; shoulder range of motion; activities of daily living (ADL); incidence of lymphedema; adherence to the interventions. Immediate outcomes after the end of VR-based interventions and delayed outcomes after a follow-up time will be included.

Search strategy

We will use a combination of text words and MeSH terms depending on the database to capture the following concepts: breast cancer, VR-based interventions. We will search the following electronic bases from their inception to May 2021: Web of Science, PubMed, Embase, Ovid, CINAHL, Cochrane Central Register of Controlled Trials, CNKI, Wanfang, VIP, and CBM. Published studies written in English and Chinese will be included. Details of search strings of the PubMed database are displayed in Table 1. The reference lists and bibliographies from relevant studies will be hand-searched for additional primary studies.

Table 1

Search strategy in PubMed

1	"Virtual Reality"[Mesh] OR VR OR “virtual reality” OR “virtual environment” OR “head-mounted display” OR "virtual reality goggle"
2	"Breast Neoplasms"[Mesh] OR “breast neoplasms” OR "breast neoplasm" OR ”breast tumors” OR “breast tumor” OR “breast cancer” OR “mammary cancer” OR “mammary cancers” OR “Breast Malignant Neoplasm” OR “Breast

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	Malignant Neoplasms" OR "Malignant Tumor of Breast" OR "Breast Malignant Tumor" OR "Breast Malignant Tumors" OR "Cancer of Breast" OR "Cancer of the Breast" OR "Mammary Carcinoma" OR "Human Mammary Carcinomas" OR "Human Mammary Carcinoma" OR "Mammary Neoplasms" OR "Human Mammary Neoplasm" OR "Human Mammary Neoplasms" OR "Mammary Neoplasm" OR "Breast Carcinoma" OR "Breast Carcinomas"
3	#1 AND #2

Data collection and analysis

Data management

Records from searches will be imported into an EndNote library (EndNote X9.1) and duplicates will be removed. We will use a piloted data collection form in Excel (Microsoft, 2003) to extract data from included studies.

Selection process

Titles and abstracts of retrieved articles will be screened against the predefined eligibility criteria by two reviewers (XB, WX) independently. We will retrieve and examine the full text of all potentially relevant articles for relevance. Any disagreement between two reviewers will be resolved by discussion or other investigators (ASKC, XL).

Data extraction

Data extraction will be carried out independently by two reviewers(XB, WX) using a pre-designed standardized form in Word (Microsoft, 2003). Any discrepancies between two reviewers will have to be resolved by discussion or other reviewers (ASKC, XL). We will remove duplicate data published in different manuscripts. Additionally, to obtain unclear or missing data, authors of the included trials will be contacted. Data extraction will include study characteristics (the first author, publication year, data collection method, study design, sampling strategy, random

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process, blinding, dropout, reporting, and others;), participant characteristics (sample size, age), intervention details (characteristics of interventions, follow-up, duration), patient-important outcomes (symptoms of pain, depression, anxiety, cognitive function, shoulder range of motion, ADL, adherence to the interventions, incidence of lymphedema).

Assessment of the risk of bias in included studies

Two reviewers will independently assess the methodological quality of all included trials. Any disagreement between two reviewers will be resolved by discussion or other investigators (ASKC, XL). The Cochrane risk-of-bias tool will be used to assess the quality of included RCTs.[29] It includes six domains of bias: selection bias, performance bias, detection bias, attrition bias, reporting bias, and other bias. Quality assessment tool of JBI evidence-based health center will be used to assess the quality of RCT or non-RCT studies. We need to select a suitable scale according to the types of studies included.

Data synthesis and statistical analysis

Data synthesis

Review Manager V.5.3 (Cochrane Collaboration) will be used to conduct this meta-analysis. The mean difference or standardized mean difference with 95% CI is used to calculate continuous variables. Forest plots will be used to display these analyze.

Assessment of heterogeneity

The I^2 test will be used to determine the statistical heterogeneity among included studies. We use a fixed-effect model for data analysis at first. If I^2 >0.5, the random-effect model will be used.[30]

Assessment of reporting biases

Publication bias will be assessed by the funnel plot method. Begg's and Egger's

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tests will be conducted to assess the publication bias. P-value<0.05 indicates the existence of publication bias.

Discussion

With the high prevalence of intense physical and psychological symptoms, function deflicts, and adverse effects in breast cancer patients during and after the treatment,[4 5] there is an urgent need for a high-quality systematic review and meta-analysis to inform evidence-based management of these problems. The proposed systematic review will report on the studies using VR-based interventions for rehabilitation management in BCS and quantitatively evaluate the efficacy of VR-based interventions in rehabilitation management of BCS. The findings of our systematic review and meta-analysis will offer BCS an alternative therapeutic option, and give directions for future researches.

So far, no systematic review or meta-analysis has focused on the VR-based interventions in rehabilitation management of the BCS. A single older review previously published described the use of VR-based interventions in cancer care, but this review only included reports published from 1993 to December 2013 in English.[26] A meta-analysis was previously published that described the effectiveness of VR-based interventions in cancer-related fatigue.[31] However, both of the studies did not retrieve enough databases, which may add the risk of bias. Additionally, they focused on cancer-related symptom management. Different types of cancer show great differences in the population, race, gender, age, and complications. Our systematic review and meta-analysis focuses on the rehabilitation of BCS, which is broader than symptom management. And we search for more electronic databases from their inception to May 2021. We will qualitatively and quantitatively examine the effects of VR-based interventions on BCS in this systematic review and meta-analysis.

Patient and public involvement

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Patient and public were not involved in the design, conduct or reporting of this study. Dissemination of findings will occur at relevant conferences or be published in a relevant journal.

Footnotes

Contributors

All authors contributed to the development of the protocol and writing of the manuscript. XB, PHFN, QC, YT, QT, XL—wrote the first draft of the manuscript. PHFN, QC, ASKC—designed the study. XB, WX, ASKC, XL—were involved in study selection, quality assessment and data extraction. XL, YT, QT will conduct the statistical analysis. All authors reviewed the manuscript, contributing critical changes and approved the final version of the manuscript.

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Competing interests

None declared.

Provenance and peer review

Not commissioned; externally peer reviewed.

Patient consent for publication

Not required.

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Reported on page #
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	3
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	11
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	11
Sponsor	5b	Provide name for the review funder and/or sponsor	11
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	11
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	4-6
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6-7
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6-7
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be	7-8

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		repeated	
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	8
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	8-9
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	7
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	9
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	N/A
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	9
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	N/A

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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Abstract

Introduction

Breast cancer is the leading cause of global cancer incidence and represents 11.7% of all new cancer cases. However, breast cancer survivors (BCSs) suffer from many intense physical and psychological symptoms, function deflicts, and adverse effects during and after the treatment, significantly affecting their quality of life. Virtual reality (VR) technology uses computer technology to create an interactive three-dimensional world by visual, audio, and touch simulation and is being used in breast cancer rehabilitation management. This paper reports on the protocol for a systematic review and meta-analysis exploring the efficacy of VR-based interventions in rehabilitation management among BCS.

Methods and analysis

This protocol for conducting a systematic review and meta-analysis was prepared according to recommendations from the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Web of Science, PubMed, Embase, CINAHL, Cochrane Central Register of Controlled Trials, CNKI, Wanfang, VIP, and SinoMed, will be used for the search. This search will include randomized controlled trials, quasi-experimental studies, and case-controlled trials published in English and Chinese. Further, the risk of bias for the studies included in the systematic review and meta-analysis will be assessed using the Cochrane risk-of-bias tool. The statistical program Review Manager V.5.3. was used for the meta-analysis. The I^2 test will be used to determine statistical heterogeneity among the included studies.

Ethics and dissemination

Ethics approval will not be needed because the data used in this systematic review and meta-analysis will be extracted from published studies. The systematic review and meta-analysis results focused on whether VR-based interventions are effective in the rehabilitation management of BCS. It will be disseminated by publication in a peer-

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reviewed journal.

Trial registration number

PROSPERO CRD42021250727

Keywords

breast cancer survivors; virtual reality; rehabilitation management; systematic review; meta-analysis; effectiveness; protocol

Strengths and limitations of this study

- To the best of our knowledge, this is the first systematic review and meta-analysis to synthesize the totality of the evidence regarding VR-based interventions in rehabilitation management of BCS.
- The results of this systematic review and meta-analysis will be helpful for clinicians in defining rehabilitation strategies to improve rehabilitation effectiveness in BCS.
- This study will be limited to English and Chinese languages, resulting in the exclusion of potentially relevant studies published in other languages.
- There is a possibility that we have a small sample size and a limited number of studies, which may influence the validity and reliability of the findings.

Word count: 1865

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Introduction

Female breast cancer has surpassed lung cancer as the leading cause of global cancer incidence in 2020, with an estimated 2.3 million new cases and 685,000 deaths, representing 11.7% of all new cancer cases and 6.9% of death cases.[1] Due to the impact of screening programs, early detection, and accessible health care services, as well as advanced therapeutic strategies, the 5-year relative survival rate for individuals with breast cancer, has increased to 82%.[2 3] However, survivors suffer from a large number of intense physical and psychological symptoms, function deflicts, and adverse effects during and after the treatment, which significantly affects their quality of life.[4 5]

Emotional distress mainly includes fatigue, pain, anxiety, and depression, common in cancer populations.[6] It was designated as the sixth vital sign in 2005 in Canada[7] and associated with a reduction in overall quality of life among cancer patients[8]. Cancer-related cognitive impairment (CRCI) is characterized as deficits in areas of cognition, including memory, attention, information processing speed, and executive function.[9 10] Between 15% and 50% of individuals with breast cancer receiving chemotherapy experience persisting cognitive impairment,[11] often referred to as “chemobrain”.[12] The duration of symptoms may extend years after the completion of treatment.[9] Due to the disruption of the lymphatic system after surgery or radiation treatment, the accumulation of lymph fluid leads to breast cancer-related lymphedema (BCRL).[13] BCRL is a chronic, potentially debilitating condition involving progressive swelling, limited range of motion, and causing feelings of pain, numbness, which requires life-long symptom management.[14] BCS are at lifelong risk for the development of BCRL.[14] The incidence is 21.4%.[15]

Virtual reality (VR) technology uses computer technology to create an interactive three-dimensional world by visual, audio, and touch simulation being used in breast cancer rehabilitation management. The development of VR offers new and noninvasive approaches for different healthcare applications. It creates interactive computer-

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generated worlds through visual, listening, and touch simulations, where an individual feels being. VR as a distraction tool improves cancer patients' emotional well-being and decreases cancer-related psychological symptoms, enhancing their quality of life.[16] In addition, VR-based rehabilitation systems (e.g., Xbox 360 Kinect games, the BrightArm Duo Rehabilitation System) have been identified to be effective for patients with weak arms and diminished grasping ability.[17] [18] These systems has been determined to be effective for patients with weak arms and diminished ability to grasp.[18] It uses VR to engage the patients in upper body bimanual exercises and simultaneously provides cognitive training and affective relief.

To the best of our knowledge, thus far, no systematic review or meta-analysis has focused on VR-based interventions in the rehabilitation management of BCS. A single older study previously published described the use of VR-based interventions in cancer care, but this review only included reports published from 1993 to December 2013 in English.[16] A previously published meta-analysis described VR-based interventions' effectiveness in cancer-related fatigue.[19] However, both studies did not retrieve enough databases, which may add the risk of bias. Additionally, they focused on the management of cancer-related symptoms. Different types of cancer show significant differences in population, race, sex, age, and complications. Our systematic review and meta-analysis focused on BCS rehabilitation, which is broader than symptom management. Therefore, we searched for more electronic databases from their inception to May 2021. In this systematic review and meta-analysis, we qualitatively and quantitatively examined the effects of VR-based interventions on BCS.

Objectives

The overarching objective of this systematic review and meta-analysis is to report on studies using VR-based interventions for rehabilitation management in BCS and to quantitatively evaluate the efficacy of VR-based interventions in breast cancer-related rehabilitation management. This will be achieved with four secondary objectives: (1) to identify the intervention type (immersive and non-immersive) that is most effective

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in breast cancer-related rehabilitation management; (2) to identify the most effective duration of intervention; and (3) to assess any lasting effect of VR-based interventions. (4) to use a simulation approach to estimate the sample size based on a meta-analysis of the existing evidence to recommend an appropriate sample size for a future trial.[20]

Methods and analysis

Study design and registration

This systematic review and meta-analysis protocol was prepared based on the recommendations of the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement[21] and registered on PROSPERO (registration number: CRD42021250727).

Eligibility criteria

Type of studies

We will include all randomized controlled trials, case-controlled trials, and quasi-experimental studies.

Types of participants/population

To obtain comprehensive studies and reduce selective bias, we will include all trials that have included patients after first or recurrent breast cancer, regardless of demographic age, race, and education status.

Types of interventions

To obtain comprehensive studies and reduce selective bias, we will select all trials assessing VR-based interventions, regardless of their aim. In trials, the selection process will not be based on the type or nature of the VR-based interventions (immersive and non-immersive virtual environments).

Types of outcome measures

Outcomes are related to rehabilitation management, such as pain, depression, anxiety, fatigue, cognitive function, shoulder range of motion, handgrip strength, lymphedema,

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cybersickness symptoms, fear of movement, bleeding, effusion, and flap necrosis after surgery.

Exclusion criteria

The exclusion criteria were as follows: (1) did not specify the type of cancer; (2) described the technologies only; and (3) were conference papers, workshop papers, literature reviews, posters, comments, letters, study protocols, or proceedings papers.

Search strategy

We will use a combination of text words and MeSH terms depending on the database to capture the following concepts: breast cancer and VR-based interventions. The following electronic databases will be searched from inception to May 2021: Web of Science, PubMed, Embase, CINAHL, Cochrane Central Register of Controlled Trials, CNKI, Wanfang, VIP, and CBM. Further, published studies written in English and Chinese were included. Details of the search strings in the PubMed database are displayed in Table 1. Furthermore, the reference lists and bibliographies from relevant studies will be manually searched for additional primary studies.

Table 1

Search strategy in PubMed

1	"Virtual Reality"[Mesh] OR VR OR "virtual reality" OR "virtual environment" OR "head-mounted display" OR "virtual reality goggle"
2	"Breast Neoplasms"[Mesh] OR "breast neoplasms" OR "breast neoplasm" OR "breast tumors" OR "breast tumor" OR "breast cancer" OR "mammary cancer" OR "mammary cancers" OR "Breast Malignant Neoplasm" OR "Breast Malignant Neoplasms" OR "Malignant Tumor of Breast" OR "Breast Malignant Tumor" OR "Breast Malignant Tumors" OR "Cancer of Breast" OR "Cancer of the Breast" OR "Mammary Carcinoma" OR "Human Mammary Carcinomas" OR "Human Mammary Carcinoma" OR "Mammary Neoplasms" OR "Human Mammary Neoplasm" OR "Human Mammary Neoplasms" OR

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	"Mammary Neoplasm" OR "Breast Carcinoma" OR "Breast Carcinomas"
3	#1 AND #2

Data collection and analysis

Data management

Records from searches will be imported into an EndNote library (EndNote X9.1), and duplicates will be removed. We will use a piloted data collection form in Excel (Microsoft, 2003) to extract data from the included studies.

Selection process

The titles and abstracts of retrieved articles will be screened independently against the predefined eligibility criteria by two reviewers (XB and WX). Moreover, we will retrieve and examine the full text of all potentially relevant articles for relevance. Any disagreement between the two reviewers will be resolved by discussion or by other investigators (ASKC, XL).

Data extraction

Data extraction will be performed independently by two reviewers (XB, WX) using a pre-designed standardized form in Word (Microsoft, 2003). Any discrepancies between the two reviewers will have to be resolved by discussion or by other reviewers (ASKC, XL). We will remove duplicate data published in different manuscripts. Additionally, to obtain unclear or missing data, the authors of the included trials will be contacted. Data extraction will consist of study characteristics (first author, publication year, data collection method, study design, sampling strategy, random process, blinding, dropout, reporting, etc.), participant characteristics (sample size, age), intervention details (characteristics of interventions, follow-up, duration), and patient-important outcomes (pain, depression, anxiety, fatigue, cognitive function, shoulder range of motion, handgrip strength, lymphedema, cybersickness symptoms, fear of movement, bleeding, effusion, and flap necrosis after surgery).

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Assessment of the risk of bias in included studies

Two reviewers will independently assess the methodological quality of all the included trials. Any disagreement between the two reviewers will be resolved by discussion or by other investigators (ASKC, XL). The Cochrane risk-of-bias tool will be used to assess the quality of included RCTs.[22] It consists of six domains of bias: selection bias, performance bias, detection bias, attrition bias, reporting bias, and other bias. The risk of bias in non-randomized studies of interventions (ROBINS-I) was used to assess the quality of the included non-RCTs, covering seven distinct domains: bias due to confounding, selection bias, bias in measurement classification of interventions, bias due to deviations from intended interventions, bias due to missing data, bias in the measurement of outcomes, and bias in the selection of the reported result.[23]

Data synthesis and statistical analysis

Data synthesis

Review Manager V.5.3 (Cochrane Collaboration) will be used to conduct the meta-analysis. The mean difference or standardized mean difference with 95% CI will be used to calculate continuous variables. Forest plots were used to perform these analyses.

Assessment of heterogeneity

The I^2 test will be used to determine statistical heterogeneity among the included studies. First, we will usefirst a fixed-effect model for data analysis. If $I^2 > 0.5$, the random-effects model will be used.[24]

Assessment of reporting biases

Publication bias will be assessed using the funnel plot method if $\geqslant 10$ included studies per outcome. Begg's and Egger's tests will be conducted to assess the publication bias. $P < 0.05$, indicating the existence of publication bias.

Discussion

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With the high prevalence of intense physical and psychological symptoms, function deflicts, and adverse effects in breast cancer patients during and after treatment,[4 5] there is an urgent need for a high-quality systematic review and meta-analysis to inform evidence-based management of these problems. The proposed systematic review will report on studies using VR-based interventions for rehabilitation management in BCS and quantitatively evaluate the efficacy of VR-based interventions in rehabilitation management of BCS. Our systematic review and meta-analysis findings will offer BCS as an alternative therapeutic option and provide directions for future research.

Patient and public involvement

Patients and the public were not involved in the design, conduct, or reporting of this study. Dissemination of findings will occur at relevant conferences or be published in an appropriate journal.

Footnotes

Contributors

All authors contributed to the development of the protocol and writing of the manuscript. XB conceived the initial idea for the study. XB, PHFN, QC, YT, QT, and XL wrote the first draft of the manuscript. ASKC and WX critically appraised the protocol and contributed to its development by revising different versions. All authors read and approved the final version of the manuscript.

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Competing interests

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None declared.

Provenance and peer review

Not commissioned; externally peer-reviewed.

Patient consent for publication

Not required.

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Reported on page #
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	3
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	10
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	10
Sponsor	5b	Provide name for the review funder and/or sponsor	10
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	10
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	4-5
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6-7
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6-7
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be	7-8

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		repeated	
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	8
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	8-9
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	7
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	9
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	9
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	N/A
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	N/A
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	9
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	N/A

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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